CH Detected

April 1, 2026

1 Project Overview: Solar Image Data Analysis

This project **automates the retrieval and processing** of solar imagery. It uses **sunpy** and **astropy** to search for specific **wavelengths** (representing different temperatures of the solar atmosphere) and **magnetic field data**, then prepares these maps for **scientific visualization**.

1.0.1 1. Library Setup and Configuration

The first step is **importing the necessary tools**. We use **numpy** for calculations, **matplotlib** for plotting, and **sunpy/astropy** for astronomical data handling.

[1]:
```
# ---------------------------
# Standard libraries
# ---------------------------
import numpy as np
import pandas as pd

# ---------------------------
# Astropy / Coordinates / Units
# ---------------------------
import astropy.units as u
from astropy.coordinates import SkyCoord

# ---------------------------
# SunPy (maps, solar constants, and data access)
# ---------------------------
import sunpy.map
from sunpy.sun import constants as sun_const
from sunpy.net import Fido
from sunpy.net import attrs as a

# ---------------------------
# Image processing
# ---------------------------
from skimage.measure import label, regionprops
from scipy.ndimage import gaussian_filter
from aiapy.calibrate import register
```

```python
# ---------------------------
# Plotting and visualization
# ---------------------------
import matplotlib.pyplot as plt
from matplotlib.colors import Normalize
from matplotlib.lines import Line2D

# ---------------------------
# Enable interactive plotting
# ---------------------------
plt.ion()
```

[1]: <contextlib.ExitStack at 0x730a54314e90>

1.0.2 2. Searching for Solar Data

We use **Fido.search** to look for specific datasets.

- **AIA Data:** Captures the **solar corona** (193 Å wavelength).
- **HMI Data:** Captures the **Line-of-Sight (LOS)** magnetic field.

1.0.3 Key Concepts for Understanding the Code

** Feature **	** Description**
Fido	The primary interface in **sunpy** to search for and download solar data from various archives.
Wavelength (193 Å)	This specific **ultraviolet wavelength** is used to observe the solar corona and hot flare plasma.
HMI LOS Field	Measures the **magnetic field strength** moving toward or away from the observer.
SunPy Map	A **2D data array** coupled with a coordinate system, allowing you to plot solar features accurately.

[2]:
```python
# Search for AIA data (Coronal intensity at 193 Angstroms)
# Time range: 2025-09-08
AIA_DATA = Fido.search(
    a.Time('2025-09-10 03:00:00', '2025-09-10 03:00:30'),
    a.Instrument.aia,
    a.Wavelength(193*u.angstrom),
    a.Physobs.intensity
)

# Search for HMI data (Magnetic field measurements)
```

```
# Time range: 2015-07-07
HMI_DATA = Fido.search(
    a.Time('2025-09-10 03:00:00', '2025-09-10 03:00:59'),
    a.Instrument.hmi,
    a.Physobs.los_magnetic_field
)
```

1.0.4 3. Fetching and Loading Maps

Once the search is complete, the files are **downloaded locally** to your **Ubuntu** environment. We then convert these files into **SunPy Map objects**, which contain both the **image data** and the **metadata** (header information).

- **Local Storage:** Files are stored as **FITS (Flexible Image Transport System)** files.
- **Map Objects:** These objects are coordinate-aware, meaning they "know" where on the Sun each pixel is located.

[3]:
```
# ---------------------------------------------------
# Define your Ubuntu download folder (ensure it exists)
# {file} will be automatically replaced by Fido with the filename
#download_path = '/home/bhairab/Documents/project/{file}'

# ---------------------------------------------------
# Fetch the data and save to the defined folder
AIA_files = Fido.fetch(AIA_DATA)#, path=download_path)
```

```
Files Downloaded:   0%|          | 0/3 [00:00<?, ?file/s]

aia.lev1.193A_2025_09_10T03_00_28.84Z.image_lev1.fits:   0%|          | 0.00/11.
↪7M [00:00<?, ?B/s]

aia.lev1.193A_2025_09_10T03_00_16.84Z.image_lev1.fits:   0%|          | 0.00/11.
↪7M [00:00<?, ?B/s]

aia.lev1.193A_2025_09_10T03_00_04.84Z.image_lev1.fits:   0%|          | 0.00/11.
↪7M [00:00<?, ?B/s]
```

[4]:
```
HMI_files = Fido.fetch(HMI_DATA)#, path=download_path)
```

```
Files Downloaded:   0%|          | 0/1 [00:00<?, ?file/s]

hmi.m_45s.2025.09.10_03_01_30_TAI.magnetogram.fits:   0%|          | 0.00/16.3M␣
↪[00:00<?, ?B/s]
```

[5]:
```
AIA_files
```

[5]:
```
<parfive.results.Results object at 0x730a3d294860>
['/home/bhairab/sunpy/data/aia.lev1.193A_2025_09_10T03_00_04.84Z.image_lev1.fits
', '/home/bhairab/sunpy/data/aia.lev1.193A_2025_09_10T03_00_16.84Z.image_lev1.fi
ts', '/home/bhairab/sunpy/data/aia.lev1.193A_2025_09_10T03_00_28.84Z.image_lev1.
fits']
```

[6]:
```
HMI_files
```

[6]:
```
<parfive.results.Results object at 0x730a3d811100>
['/home/bhairab/sunpy/data/hmi.m_45s.2025.09.10_03_01_30_TAI.magnetogram.fits']
```

[7]:
```
# Create SunPy Map objects from the downloaded FITS files
# aia_193: Representative of the hot solar corona
aia193 = sunpy.map.Map(AIA_files[0])

# hmi6173: Representative of the magnetic photosphere
hmi6173 = sunpy.map.Map(HMI_files[0])
```

1.0.5 4. Aligning Different Solar Perspectives

In solar physics, different instruments (like **AIA** and **HMI**) have different **pixel scales**, **resolutions**, and **physical viewing properties**. Furthermore, these instruments operate on different **cadences**:

- **AIA (Atmospheric Imaging Assembly):** Captures a new image roughly every **12 seconds**, providing high-speed monitoring of the **solar corona**.
- **HMI (Helioseismic and Magnetic Imager):** Captures data roughly every **45 seconds**, focusing on the **Sun's magnetic field** and surface.

To compare them **pixel-for-pixel**—for example, to see exactly which **magnetic feature** on the surface (**HMI**) is causing a bright flare in the corona (**AIA**)—the images must be **perfectly aligned**. Because AIA images are updated nearly four times faster than HMI, we must **reproject** the high-cadence AIA data to match the **coordinate system (WCS)** of the HMI master frame.

[8]:
```
# ----------------------------------------------------
# ALIGNMENT: Reprojecting AIA to match HMI coordinates
# ----------------------------------------------------
aia_reprojected = aia193.reproject_to(hmi6173.wcs)
aia_reprojected.nickname = 'AIA 193 Å (Reprojected)'
```

```
2026-04-01 07:13:19 - reproject.common - INFO: Calling _reproject_full in non-
dask mode
```

1.0.6 5. Comparative Visualization of Aligned Solar Data

In solar physics, we often need to see how the **magnetic "roots"** on the Sun's surface (**photosphere**) relate to the **hot plasma structures** above them (**corona**).

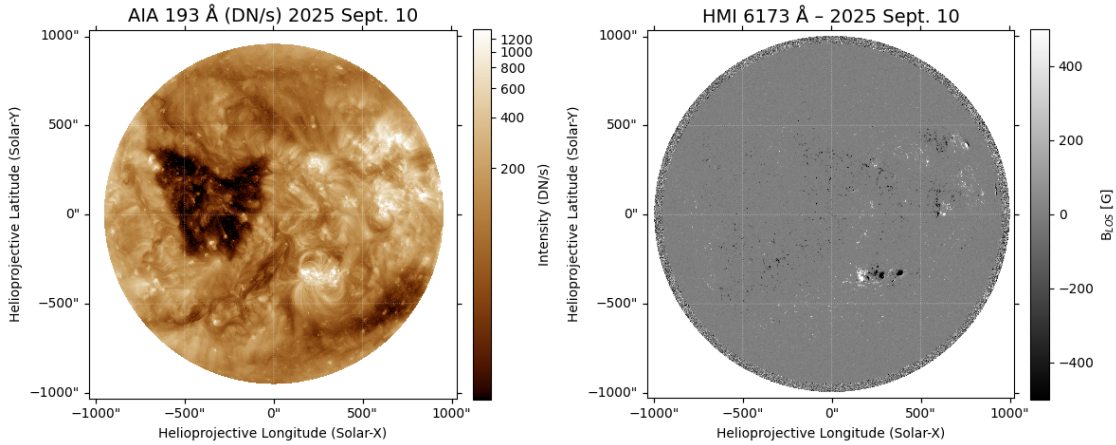
- **AIA (12s cadence):** Shows the **high-temperature corona**. Because it is **reprojected**, every pixel now matches the **HMI grid**.
- **HMI (45s cadence):** Shows the **magnetic field strength** (B_{LOS}). The **grayscale map** helps distinguish between **positive (white)** and **negative (black)** magnetic polarities.

1.0.7 Visualizing the Connection

By aligning these two data sources, we can identify how **active regions** in the corona are fueled by underlying **magnetic flux**. This comparative view is essential for predicting **solar flares** and understanding the energy transfer between different layers of the solar atmosphere.

[9]:
```
# ====================================================
# PROCESSING: Normalize AIA 193 Å by Exposure Time
# ====================================================
# Use exposure time from the ORIGINAL AIA map (not reprojected)
aia_exptime = aia193.exposure_time.to_value()      # seconds

# Normalize reprojected map using original exposure time
aia_norm = aia_reprojected / aia_exptime           # DN/s


# ====================================================
# FIGURE SETUP
# ====================================================
fig = plt.figure(figsize=(12, 5))


# ====================================================
# PANEL 1: AIA 193 Å (Exposure-Normalized, AIA Colormap)
# ====================================================
ax1 = fig.add_subplot(121, projection=aia_reprojected)

im1 = aia_norm.plot(
    axes=ax1,
    cmap="sdoaia193",
    clip_interval=(1, 99.9) * u.percent
)

ax1.set_title(
    "AIA 193 Å (DN/s) 2025 Sept. 10",
    fontsize=14
)

cb1 = fig.colorbar(
    im1,
    ax=ax1,
    orientation='vertical',
    fraction=0.046,
    pad=0.04
)
cb1.set_label("Intensity (DN/s)")

ax1.invert_xaxis()
```

```python
ax1.invert_yaxis()


# ======================================================
# PANEL 2: HMI 6173 Å (LOS Magnetic Field, Grayscale)
# ======================================================
ax2 = fig.add_subplot(122, projection=hmi6173)

im2 = hmi6173.plot(
    axes=ax2,
    cmap="gray",
    norm=Normalize(-500, 500)
)

ax2.set_title(
    "HMI 6173 Å - 2025 Sept. 10",
    fontsize=14
)

cb2 = fig.colorbar(
    im2,
    ax=ax2,
    orientation='vertical',
    fraction=0.046,
    pad=0.04
)
cb2.set_label("B$_{LOS}$ [G]")

ax2.invert_xaxis()
ax2.invert_yaxis()


# ======================================================
# FINAL LAYOUT & EXPORT
# ======================================================
plt.tight_layout(pad=1.2)

plt.savefig(
    "/home/bhairab/Documents/project/reproject.png",
    dpi=600,
    bbox_inches="tight"
)

plt.show()
```



1.0.8 6. Region of Interest (ROI) and Advanced Visualization

To move beyond full-disk observations, we focus on specific solar features—such as **flares** or **coronal holes**—by defining a precise **Region of Interest (ROI)** and applying advanced scaling techniques.

Defining a Region of Interest (ROI) To study a specific feature, we define a "box" in space using **Helioprojective Coordinates**, which measure distances in **arcseconds (arcsec)** from the center of the Sun as seen from Earth.

- **SkyCoord:** This creates specific mathematical points in the sky based on solar metadata.
- **Bottom-left & Top-right:** These coordinates define the exact boundaries of the area we want to **"zoom in"** on.

Visualizing the Selected Region This step plots the full Sun but overlays a **dashed box** over the selected area. This is a crucial step for scientific publications to show exactly where the data was sampled.

- **draw_quadrangle:** Automatically draws the box based on the **SkyCoord** points defined above.
- **clip_interval:** This adjusts the contrast by ignoring the brightest **0.1%** and darkest **1%** of pixels, making subtle details of the solar atmosphere more visible to the human eye.

Data Normalization (DN/s) Raw satellite data is often recorded in **"Data Numbers" (DN)**. However, because the exposure time (how long the camera shutter is open) can vary, we must divide the pixel values by the exposure time.

- **Standardization:** This converts the image from **"total light"** to **"light per second" (DN/s)**, allowing for a fair comparison between images taken at different times or with different instruments.

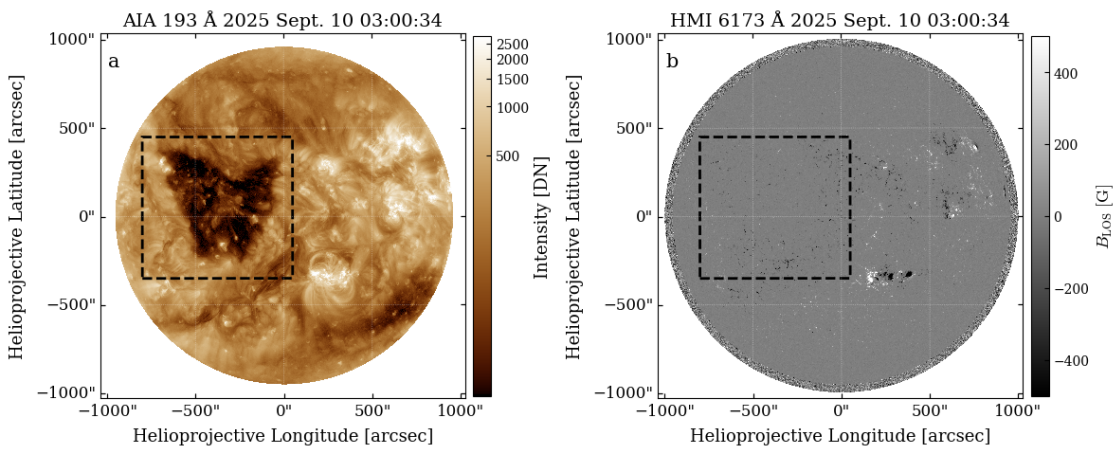
Final Comparison and Export The final block of the project creates a **high-resolution side-by-side comparison** of the **Coronal atmosphere (AIA)** and the **Magnetic surface (HMI)**.

- **Invert Axis:** Ensures the orientation matches standard **solar North/South** maps.
- **savefig:** Exports the final result as a high-quality **.png** file (**600 DPI**) suitable for a thesis or research paper.

[16]:
```
# ---------------------------------------------------
# Define corners in the AIA coordinate frame
# ---------------------------------------------------
aia_bottom_left = SkyCoord(-800 * u.arcsec, -350 * u.arcsec,
                           frame=aia_reprojected.coordinate_frame)
aia_top_right   = SkyCoord(50 * u.arcsec,  450 * u.arcsec,
                           frame=aia_reprojected.coordinate_frame)

# ---------------------------------------------------
# Create figure
# ---------------------------------------------------
fig = plt.figure(figsize=(12, 6))

# ===================== AIA Plot =====================
ax1 = fig.add_subplot(121, projection=aia_reprojected.wcs)
im1 = aia_reprojected.plot(
    axes=ax1,
    clip_interval=(1, 99.9) * u.percent
)

aia_reprojected.draw_quadrangle(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right,
    axes=ax1,
    edgecolor="black",
    linestyle="--",
    linewidth=2
)

ax1.set_title("AIA 193 Å 2025 Sept. 10 03:00:34", fontsize=14)

ax1.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)

# Enforce uniform label rendering (Astropy-safe)
ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

ax1.tick_params(labelsize=12)
ax1.invert_xaxis()
ax1.invert_yaxis()
```

```python
cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label("Intensity [DN]", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

ax1.text(
    0.02, 0.95, "a",
    transform=ax1.transAxes,
    va="top", ha="left",
    fontsize=15
)

# ===================== HMI Plot =====================
ax2 = fig.add_subplot(122, projection=hmi6173.wcs)
im2 = hmi6173.plot(
    axes=ax2,
    cmap="grey",
    norm=Normalize(-500, 500)
)

hmi6173.draw_quadrangle(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right,
    axes=ax2,
    edgecolor="black",
    linestyle="--",
    linewidth=2
)

ax2.set_title("HMI 6173 Å 2025 Sept. 10 03:00:34", fontsize=14)

ax2.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)

# Enforce uniform label rendering (Astropy-safe)
ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

ax2.tick_params(labelsize=12)
ax2.invert_xaxis()
ax2.invert_yaxis()

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}}\ \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)

ax2.text(
```

```
    0.02, 0.95, "b",
    transform=ax2.transAxes,
    va="top", ha="left",
    fontsize=15
)

# ----------------------------------------------------
# Layout + Save
# ----------------------------------------------------
plt.tight_layout(pad=1.2)

plt.savefig(
    "/home/bhairab/Documents/project/chbox.png",
    dpi=600,
    bbox_inches="tight"
)

plt.show()
```



1.0.9 7. Extracting and Formatting Submaps for Publication

In this stage, we **"crop"** the large solar images to focus exclusively on the area defined by your **coordinate box**. This allows for a much higher level of detail when analyzing local solar features like **coronal holes** or **active regions**.

- **Submap Creation:** Using the **SkyCoord** boundaries defined earlier, we slice the full-disk map into a smaller, high-resolution **submap**.
- **Coordinate Preservation:** Unlike a standard image crop, a **SunPy submap** retains all the **WCS (World Coordinate System)** metadata, so your axes still show correct **arcsecond** values.
- **Publication-Ready Styling:** We apply specific **color maps** (like `sdoaia193`) and adjust the **stretch** (logarithmic or linear) to ensure that the solar features are visually distinct for

research papers.

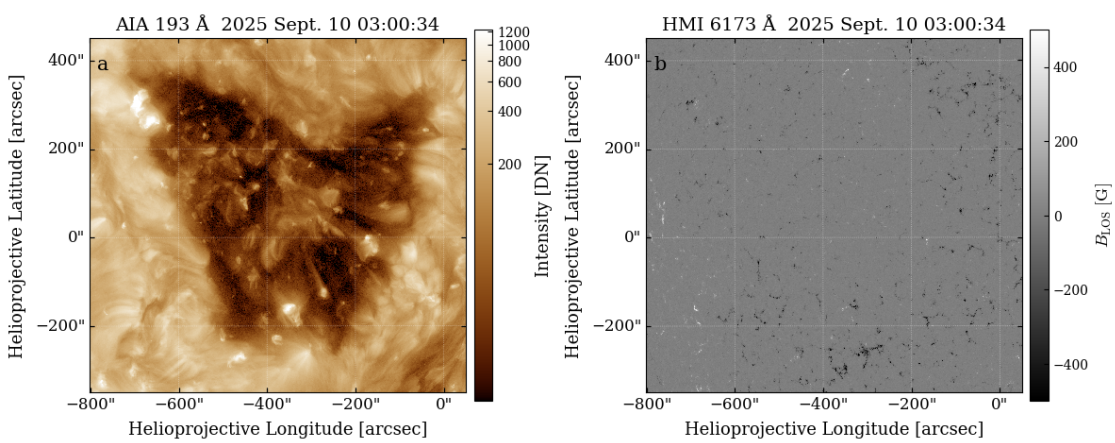
1.0.10 Key Benefits of Submapping:

1. **Reduced File Size:** Working with cropped data is computationally faster for time-series analysis.
2. **Feature Centering:** You can track a specific **Active Region (AR)** as it rotates across the solar disk.
3. **Enhanced Detail:** By removing the "black space" around the Sun, the **normalization** and **scaling** algorithms focus only on the relevant solar plasma.

[17]:
```
# ------------------------------------------------
# Global ApJ-safe style (ONE place)
# ------------------------------------------------
plt.rcParams.update({
    "font.family": "serif",
    "mathtext.fontset": "cm",
    "xtick.direction": "in",
    "ytick.direction": "in",
})

# ------------------------------------------------
# Create submaps from defined region
# ------------------------------------------------
aia_sub  = aia_reprojected.submap(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right
)
hmi_sub1 = hmi6173.submap(
    bottom_left=aia_bottom_left,
    top_right=aia_top_right
)

# ------------------------------------------------
# Create figure
# ------------------------------------------------
fig = plt.figure(figsize=(12, 6))

# ===================== AIA submap =====================
ax1 = fig.add_subplot(121, projection=aia_sub.wcs)
im1 = aia_sub.plot(
    axes=ax1,
    clip_interval=(1, 99.9) * u.percent
)

ax1.set_title("AIA 193 Å  2025 Sept. 10 03:00:34", fontsize=14)
```

```
ax1.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)

# Enforce uniform label rendering (Astropy-safe)
ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

ax1.tick_params(labelsize=12)
ax1.invert_xaxis()
ax1.invert_yaxis()

cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label("Intensity [DN]", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

ax1.text(
    0.02, 0.95, "a",
    transform=ax1.transAxes,
    va="top", ha="left",
    fontsize=15
)

# ===================== HMI submap =====================
ax2 = fig.add_subplot(122, projection=hmi_sub1.wcs)
im2 = hmi_sub1.plot(
    axes=ax2,
    cmap="grey",
    norm=Normalize(-500, 500)
)

ax2.set_title("HMI 6173 Å  2025 Sept. 10 03:00:34", fontsize=14)

ax2.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)

# Enforce uniform label rendering (Astropy-safe)
ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

ax2.tick_params(labelsize=12)
ax2.invert_xaxis()
ax2.invert_yaxis()

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}}\ \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)
```

```
ax2.text(
    0.02, 0.95, "b",
    transform=ax2.transAxes,
    va="top", ha="left",
    fontsize=15
)

# ---------------------------------------------------
# Final layout and show/save
# ---------------------------------------------------
plt.tight_layout(pad=1.0)

plt.savefig(
    "/home/bhairab/Documents/project/chonly.png",
    dpi=600,
    bbox_inches="tight"
)

plt.show()
```



1.0.11 8. Feature Extraction and Boundary Mapping

Instead of just looking at the Sun, we are now asking the computer to **"find"** a dark region (**Coronal Hole**) and draw a **boundary** around it. We then overlay this boundary on the **HMI magnetic map** to see what the magnetic field looks like specifically inside that region.

---

a. Segmentation & Masking The code identifies **"dark" pixels** (low intensity) which usually indicate **Coronal Holes** in **AIA 193 Å** images.

- **Thresholding:** It flags any pixel with an intensity lower than **100** as "interesting."

- **Gaussian Smoothing:** It applies a slight **"blur"** to the detection. This removes **"noise"** (tiny random dots) so you get a smooth, continuous shape instead of a jagged one.
- **Labeling:** It identifies separate **"islands"** of dark pixels. The code is programmed to pick the **largest island** (the main Coronal Hole).

b. Identifying the Centroid The code calculates the exact **center (centroid)** of the detected hole. This is useful for labeling the feature with a **marker** (in this case, "0") to track its movement over time.

c. Plotting with Contours (Panel A & B) The most important visual tool here is `ax.contour`.

- **Outline Detection:** It takes the **"outline"** found in the AIA image and draws it as a **red line**.
- **Cross-Instrument Overlay:** Because your images are **reprojected**, the red line is drawn in the **exact same physical location** on both the **AIA plasma map** and the **HMI magnetic map**.

1.0.12 Scientific Insight

By placing the **AIA contour** onto the **HMI magnetogram**, researchers can confirm if the "dark hole" corresponds to an **"open" magnetic field line**—the source of high-speed solar wind that can affect Earth's magnetosphere.

1.0.13 Important Terms to Know

To understand the final visualization and analysis, it is helpful to define the technical parameters used to process the solar imagery.

Term	Meaning in this Code
Gaussian Filter	A mathematical tool to **"smooth"** data, removing speckle noise from the detection.
Sigma (σ)	The strength of the smoothing. A **higher sigma** makes the boundary smoother but less detailed.
Centroid	The **geometric center** of the detected region, used for tracking and labeling.
Contour	The **red line** representing the **"edge"** or **"boundary"** of the solar feature.
GridSpec	A layout tool that allows **precise control** over how the two plots are positioned side-by-side.

1.0.14 Visualizing the Process

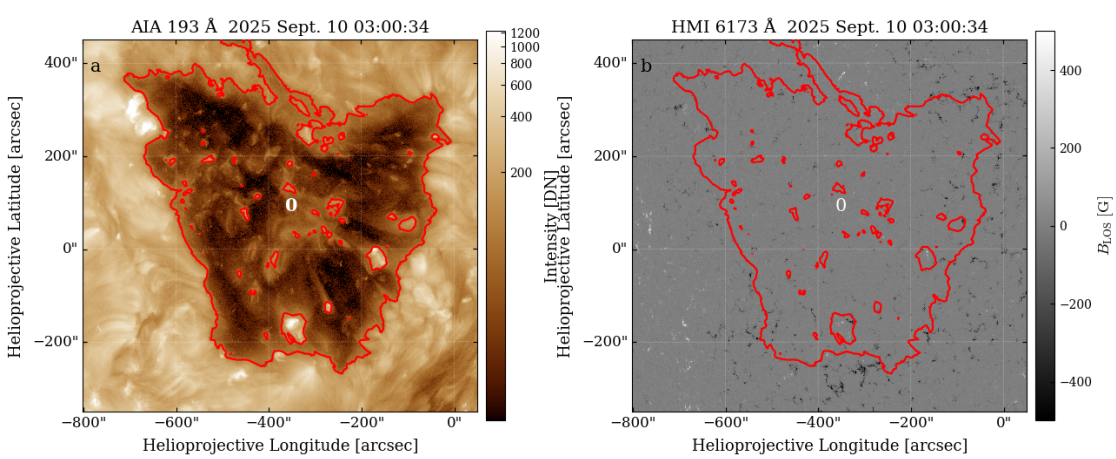
When we apply these terms, we are essentially turning a "fuzzy" satellite image into a **mathematical object**.

- The **Gaussian Filter** (σ) ensures that we don't accidentally label a single dark pixel as a "hole."
- The **Contour** allows us to export that shape and overlay it onto the **HMI magnetic map** to see if the magnetic field is "open" or "closed" within that specific boundary.

[18]:
```
# Global ApJ-uniform style (SAFE) - same as above
# ------------------------------------------------
plt.rcParams.update({
    "font.family": "serif",
    "mathtext.fontset": "cm",
    "xtick.direction": "in",
    "ytick.direction": "in",
})

# ------------------------------------------------
# Segment AIA dark regions + smooth
# ------------------------------------------------
threshold = 100
segmented = aia_sub.data < threshold
smoothed_mask = gaussian_filter(segmented.astype(float), sigma=2) > 0.5

# Label regions (use AIA intensity image for region finding)
labeled = label(smoothed_mask)
regions = regionprops(labeled, intensity_image=aia_sub.data)
regions = sorted(regions, key=lambda r: r.area, reverse=True)

# Safety: handle empty detection
if len(regions) == 0:
    raise ValueError("No regions found. Try adjusting threshold/sigma.")

# Keep only largest region
largest_region = regions[0]
single_mask = (labeled == largest_region.label)

# Region centroid (pixel)
cy, cx = largest_region.centroid  # (row, col)

# ------------------------------------------------
# Plot (UNIFORM with the earlier figure: GridSpec + per-axis colorbars)
# ------------------------------------------------
fig = plt.figure(figsize=(13, 6))
gs = fig.add_gridspec(1, 2)

# ===================== AIA panel =====================
ax1 = fig.add_subplot(gs[0, 0], projection=aia_sub.wcs)
im1 = aia_sub.plot(axes=ax1, clip_interval=(1, 99.9) * u.percent)
ax1.contour(single_mask, levels=[0.5], colors="red", origin="lower")
```

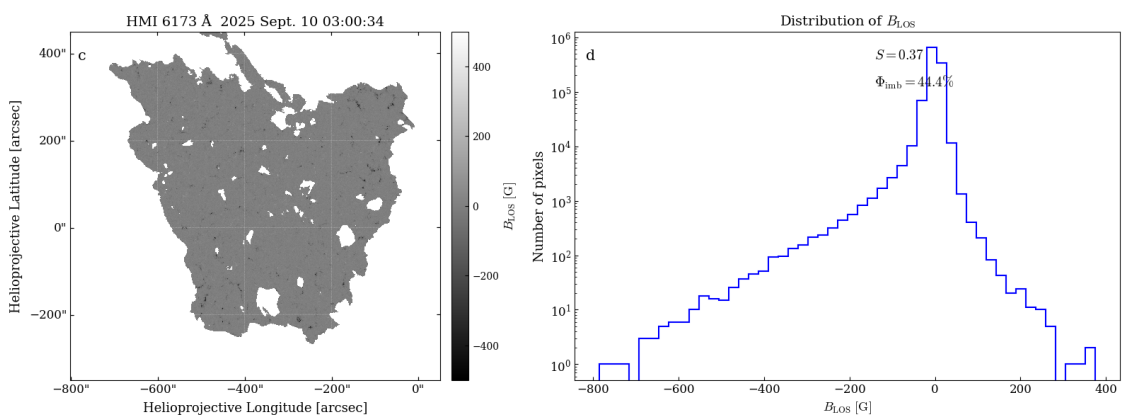
```
ax1.set_title("AIA 193 Å  2025 Sept. 10 03:00:34", fontsize=14)

ax1.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

# WCS tick labels (PDF consistency)
ax1.coords[0].set_ticklabel(size=12)
ax1.coords[1].set_ticklabel(size=12)

# label at centroid (bold like your other labels)
ax1.text(cx, cy, "0", color="white",
         ha="center", va="center", fontsize=15, fontweight="bold")

# panel label
ax1.text(0.02, 0.95, "a",
         transform=ax1.transAxes, va="top", ha="left",
         fontsize=15, color="black")

ax1.invert_xaxis()
ax1.invert_yaxis()

cbar1 = fig.colorbar(im1, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label("Intensity [DN]", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

# ===================== HMI panel =====================
ax2 = fig.add_subplot(gs[0, 1], projection=hmi_sub1.wcs)
im2 = hmi_sub1.plot(axes=ax2, cmap="grey", norm=Normalize(-500, 500))
ax2.contour(single_mask, levels=[0.5], colors="red", origin="lower")

ax2.set_title("HMI 6173 Å  2025 Sept. 10 03:00:34", fontsize=14)

ax2.coords[0].set_axislabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.coords[1].set_axislabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax2.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax2.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)

# WCS tick labels (PDF consistency)
ax2.coords[0].set_ticklabel(size=12)
ax2.coords[1].set_ticklabel(size=12)

# same centroid label
ax2.text(cx, cy, "0", color="white",
         ha="center", va="center", fontsize=15)
```

```python
# panel label
ax2.text(0.02, 0.95, "b",
         transform=ax2.transAxes, va="top", ha="left",
         fontsize=15, color="black")

ax2.invert_xaxis()
ax2.invert_yaxis()

cbar2 = fig.colorbar(im2, ax=ax2, fraction=0.046, pad=0.02)
cbar2.set_label(r"$B_{\mathrm{LOS}}\ \mathrm{[G]}$", fontsize=13)
cbar2.ax.tick_params(labelsize=10)

# ---------------------------------------------------
# IMPORTANT: no tight_layout for WCSAxes + PDF
# ---------------------------------------------------

# ---------------------------------------------------
# Layout + Save
# ---------------------------------------------------
plt.subplots_adjust(left=0.07, right=0.93, bottom=0.10, top=0.90, wspace=0.30)

plt.savefig(
    "/home/bhairab/Documents/project/ch_boundary.png",
    dpi=600,
    bbox_inches="tight"
)

plt.show()
```



[20]:
```
# Create mask from largest region
bbox = regions[0].bbox
mask = np.ones_like(hmi_sub1.data, dtype=bool)
mask[bbox[0]: bbox[2], bbox[1]: bbox[3]] = ~regions[0].image

# Create masked SunPy map
hmi_masked = sunpy.map.Map((hmi_sub1.data, hmi_sub1.meta), mask=mask)

# Extract unmasked data
unmasked_data = hmi_masked.data[~hmi_masked.mask]

# Create figure with two subplots
fig = plt.figure(figsize=(18, 6))

# ===================== Masked HMI Map =====================
ax1 = fig.add_subplot(121, projection=hmi_masked)
im = hmi_masked.plot(axes=ax1, cmap="grey", norm=Normalize(-500, 500))

ax1.set_title("HMI 6173 Å  2025 Sept. 10 03:00:34", fontsize=14)
ax1.set_xlabel("Helioprojective Longitude [arcsec]", fontsize=13)
ax1.set_ylabel("Helioprojective Latitude [arcsec]", fontsize=13)
ax1.tick_params(labelsize=12, direction="in")

ax1.invert_xaxis()
ax1.invert_yaxis()

#  Colorbar EXACTLY like the AIA+HMI figure (fraction + pad only)
cbar1 = fig.colorbar(im, ax=ax1, fraction=0.046, pad=0.02)
cbar1.set_label(r"$B_{\mathrm{LOS}}\ \mathrm{[G]}$", fontsize=13)
cbar1.ax.tick_params(labelsize=10)

# Panel label
ax1.text(0.02, 0.95, "c",
         transform=ax1.transAxes,
         va="top", ha="left",
         fontsize=15)

# ===================== Histogram =====================
ax2 = fig.add_subplot(122)
ax2.hist(unmasked_data, bins=50, histtype="step", color="blue", linewidth=1.5)

ax2.set_yscale("log")
ax2.set_xlabel(r"$B_{\mathrm{LOS}}\ \mathrm{[G]}$", fontsize=13)
ax2.set_ylabel("Number of pixels", fontsize=13)
ax2.set_title(r"Distribution of $B_{\mathrm{LOS}}$", fontsize=14)
ax2.tick_params(labelsize=12, direction="in")
```

```
# Annotations
ax2.text(0.55, 0.95, r"$S = 0.37$",
         transform=ax2.transAxes,
         va="top", ha="left",
         fontsize=14)

ax2.text(0.55, 0.88, r"$\Phi_{\mathrm{imb}} = 44.4\%$",
         transform=ax2.transAxes,
         va="top", ha="left",
         fontsize=14)

# Panel label
ax2.text(0.02, 0.95, "d",
         transform=ax2.transAxes,
         va="top", ha="left",
         fontsize=15)

# Layout + Save
# ----------------------------------------------------
plt.tight_layout(pad=1.2)
plt.savefig(
    "/home/bhairab/Documents/project/Coronal_map_histogram1.png",
    dpi=600,
    bbox_inches="tight"
)
plt.show()
```



1.0.15 9. Calculating Physical Properties and Magnetic Flux

In this stage, we convert the pixels observed in solar images into **physical dimensions** on the Sun's surface. Using solar constants, such as the Sun's radius, we translate angular measurements from Earth ("arcseconds") into physical units like **centimeters and meters**. This allows us to quantify the area of **Coronal Holes (CHs)** and calculate their **total magnetic flux**.

1.0.16 Step-by-Step Breakdown

a. Physical Area Calculation The Sun is approximately **150 million kilometers** away, so a single pixel on the camera corresponds to a large area on the solar surface. The process involves: * **Pixel Scale:** Determining how many **arcseconds** each pixel covers. * **Geometry Conversion:** Converting these arcseconds to radians and multiplying by the Sun's radius ($R_\odot$) to find the physical length and width of a pixel in centimeters. * **Pixel Area:** Multiplying the length and width yields the area of a single pixel in cm^2.

b. Coronal Hole Area The total area of a coronal hole is calculated as the number of detected pixels (N_{pixels}) multiplied by the area of a single pixel. The area is typically expressed in: * m^2: Standard square meters. * Mm^2: Megameters squared (1 Mm = 1,000 km), the standard unit in solar physics for large-scale features.

c. Total Unsigned Magnetic Flux (Φ or Φ_{uns}) This step calculates the **total unsigned magnetic flux** within the CH boundary. * **Absolute Value:** The absolute value ($|B_{\text{LOS}}|$) of each magnetic pixel is taken to ensure that positive and negative magnetic fields do not cancel each other out. * **Summation:** These absolute values are summed and multiplied by the pixel area. * **Units:** The resulting flux is measured in **Maxwell (Mx)**, representing the total number of magnetic field lines passing through the Coronal Hole.

1.0.17 Mathematical Formulation

Following the approach of **Hofmeister et al. (2017)**, the total unsigned magnetic flux is defined as:

$$\Phi = \int_A |B_{\text{LOS}}| \, dA \approx \sum_i |B_i| \, A_i(\mu)$$

Where: * $|B_i|$ is the absolute value of the **Line-of-Sight (LOS)** magnetic field in the i-th pixel. Taking the absolute value ensures the total magnetic "energy" is measured without cancellation between polarities. * $A_i(\mu)$ is the area of the i-th pixel, corrected for the **heliocentric angle** θ. * $\mu = \cos\theta$ is the cosine of the angle between the observer and the solar surface (correcting for the spherical curvature of the Sun).

1.0.18 References

Hofmeister, S. J., Veronig, A., Reiss, M. A., Temmer, M., Vennerstrom, S., Vršnak, B., & Heber, B. (2017). Characteristics of low-latitude coronal holes near the maximum of solar cycle 24. The Astrophysical Journal, 835(2), 268.

[23]:
```
# ---- AIA: average intensity inside CH mask (DN)
aia_avg_intensity = np.mean(aia_sub.data[single_mask])
```

```python
# ---- Number of pixels in CH
N_pixels = single_mask.sum()  # Only CH pixels!

# ---- Pixel scale (arcsec / pixel)
scale_x = hmi_masked.scale[0].to_value(u.arcsec / u.pixel)
scale_y = hmi_masked.scale[1].to_value(u.arcsec / u.pixel)

# ---- Constants
arcsec_to_rad = (1 * u.arcsec).to_value(u.rad)
R_sun_cm = sun_const.radius.to_value(u.cm)

# ---- Pixel area (cm^2)
pixel_area_cm2 = (scale_x * arcsec_to_rad * R_sun_cm) * \
                 (scale_y * arcsec_to_rad * R_sun_cm)

# ---- Coronal hole area
ch_area_cm2 = N_pixels * pixel_area_cm2
ch_area_m2 = ch_area_cm2 * 1e-4
ch_area_Mm2 = ch_area_cm2 * 1e-16

# ---- Total unsigned magnetic flux (Mx)
total_unsigned_flux = np.sum(
    np.abs(hmi_sub1.data[single_mask]) * pixel_area_cm2
)

# ---- Print results
print(f"Observation datetime        = {observation_datetime}")
print("====== Coronal Hole Properties ======")
print(f"Average AIA intensity        = {aia_avg_intensity:.2f} DN")
print(f"Total unsigned magnetic flux = {total_unsigned_flux:.2e} Mx")
print(f"Coronal Hole area            = {ch_area_m2:.2e} m^2")
print(f"Coronal Hole area            = {ch_area_Mm2:.2f} Mm^2")
```

```
Observation datetime        = 2025 Sept. 10 03:00:34
====== Coronal Hole Properties ======
Average AIA intensity        = 47.61 DN
Total unsigned magnetic flux = 3.58e+17 Mx
Coronal Hole area            = 3.16e+12 m^2
Coronal Hole area            = 3.16 Mm^2
```